

Parallel Tether Control System for Data Collection in Nuclear Disaster Site

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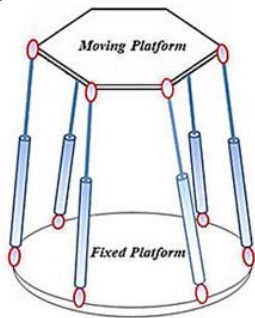
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Problem definition

- Collecting data in a nuclear disaster site is challenging due to high risk-level for human.
- It is essential to collect data not only ground level but also higher places.
- Ground robots cannot reach to higher places in the site.
- Drones are better solution but, has some limitations.
- Limitations of drones
 - Communication limitations
 - Restricted flight time and range
 - Air turbulence generated by propellers

Motivation

- Similar approach (RESQ-A) has been developed by JAEA (scissor lift).
- **Parallel link manipulator** with 6 DoF provide better control of position and orientation of the payload.
- Ground robots have better stability, long running time, and robustness over drones.
- Helium balloon can be used to provide upward force to move a module that comprises with multiple sensors (payload) to higher position.



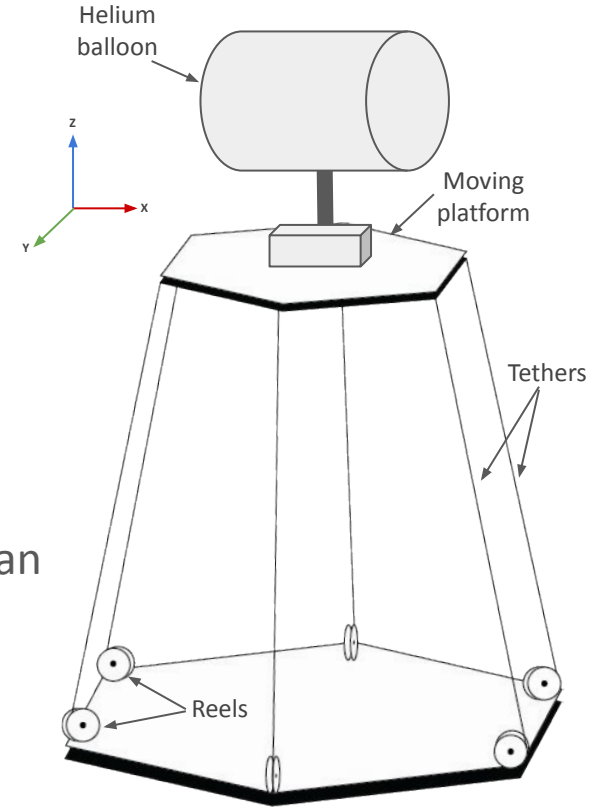
Source:
<https://www.fujipress.jp/jrm/rb/robot002800030404/>



RESQ-A robots (Source: <https://doi.org/10.1186/s40648-017-0073-7>)

Research goal

- Main goal is to control position and orientation of moving platform in all X, Y, Z axes.
- Maximum elevation has to be 3m.
- The moving platform attached to a helium balloon is controlled by 6 tethers.
- Each tether control module (reel) comprises a motor, an encoder, and a processing board.
- Whole system is moved by crawlers to the targeted disaster site (not developed yet).



Sketch of the parallel tether system

Methodology

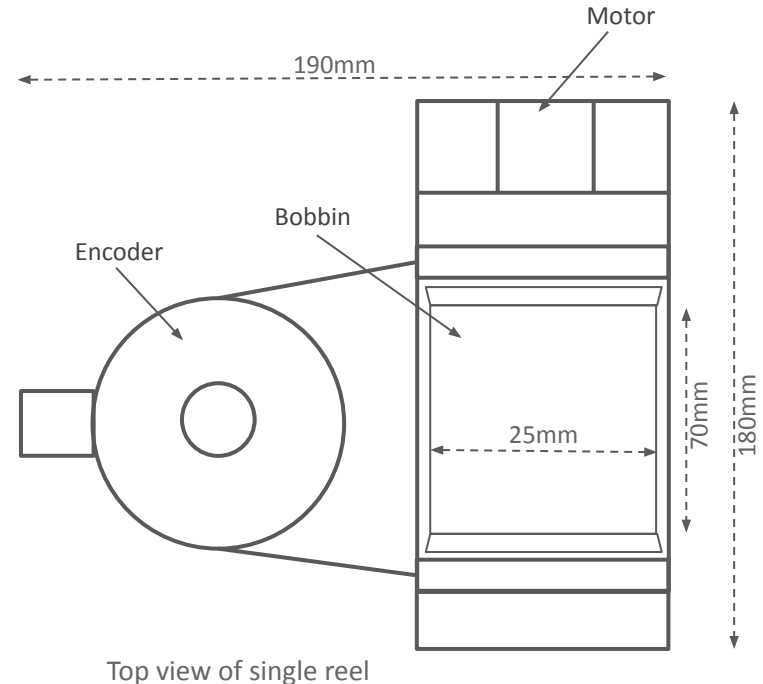
- Position and orientation of the moving platform are changed by changing the **length** of the tether.
- Given length is converted into number of turns the bobbin has be turned.

$$n_{revs} = \frac{L}{2\pi r} + \alpha$$

- Number of turns is been sent to the motor to pull or push the tether.
- Rotation around X and Y axes can be achieved pulling 3 tethers perpendicular to the corresponding axis.
- It is impossible to rotate the sensor module around z axis, but it can be achieved by rotating the whole system.

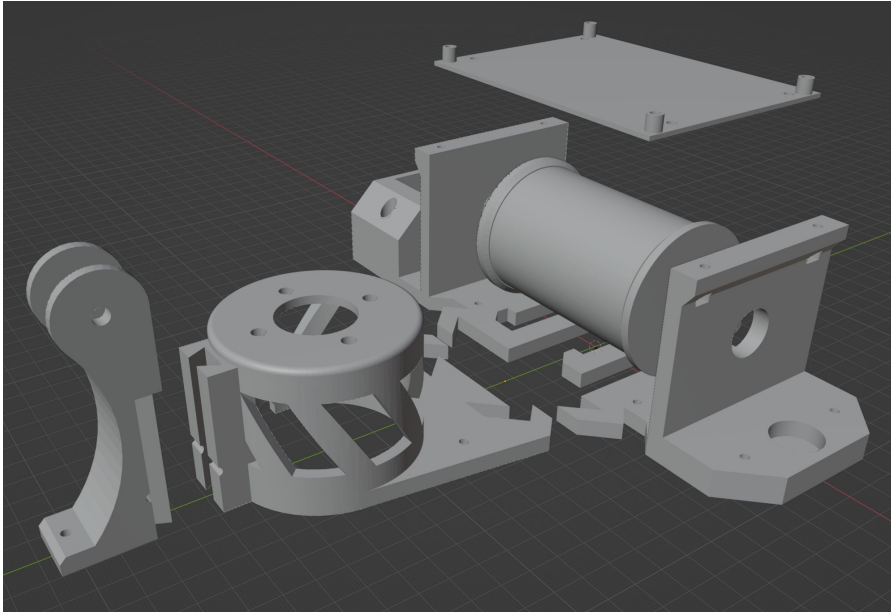
Development of a single reel

- Dynamixel (XL430-W250-T) powered reel with max torque 1.4Nm
- Open CR processing board
 - Input voltage: 12V
 - Communication interface: UART
- Bobbin radius: 25mm
- Bobbin length: 70mm
- Dimensions of a single reel:
 - Width: 180mm
 - Length: 190mm
 - Height: 100mm

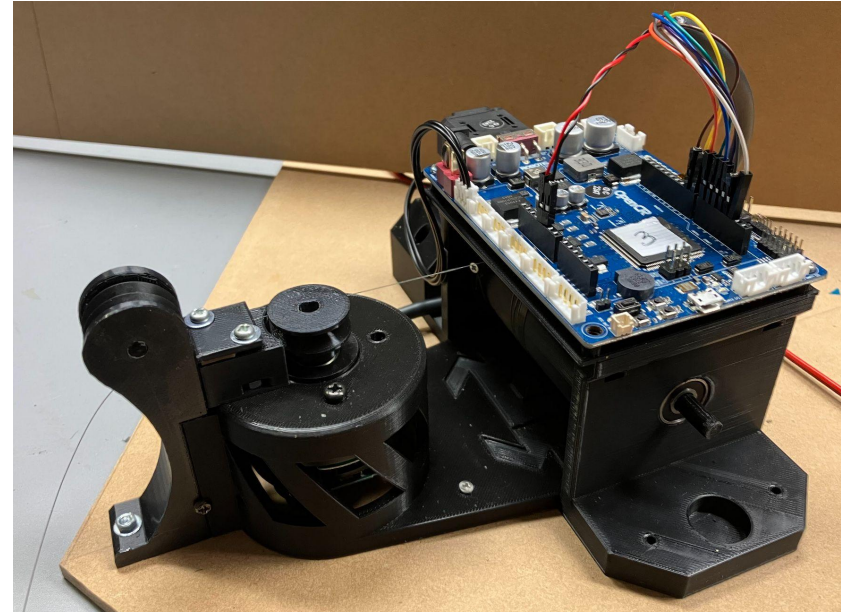


3D Assembly of single reel

- 3D printable parts were designed using FreeCAD software.
- Accuracy of a reel has been tested before create complete assembly.



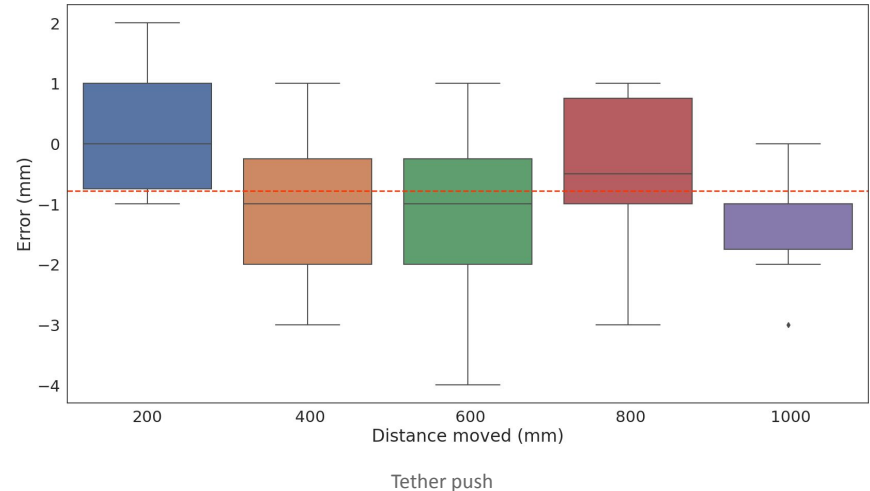
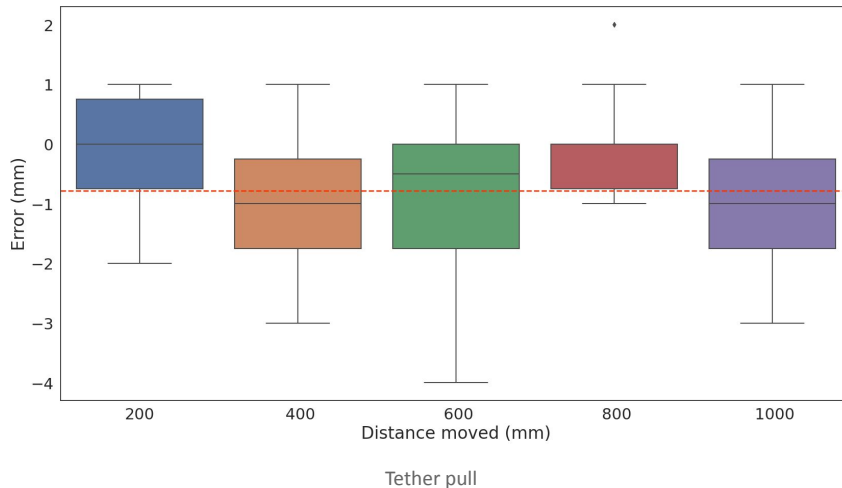
Disassembly of single reel



3D printed assembly

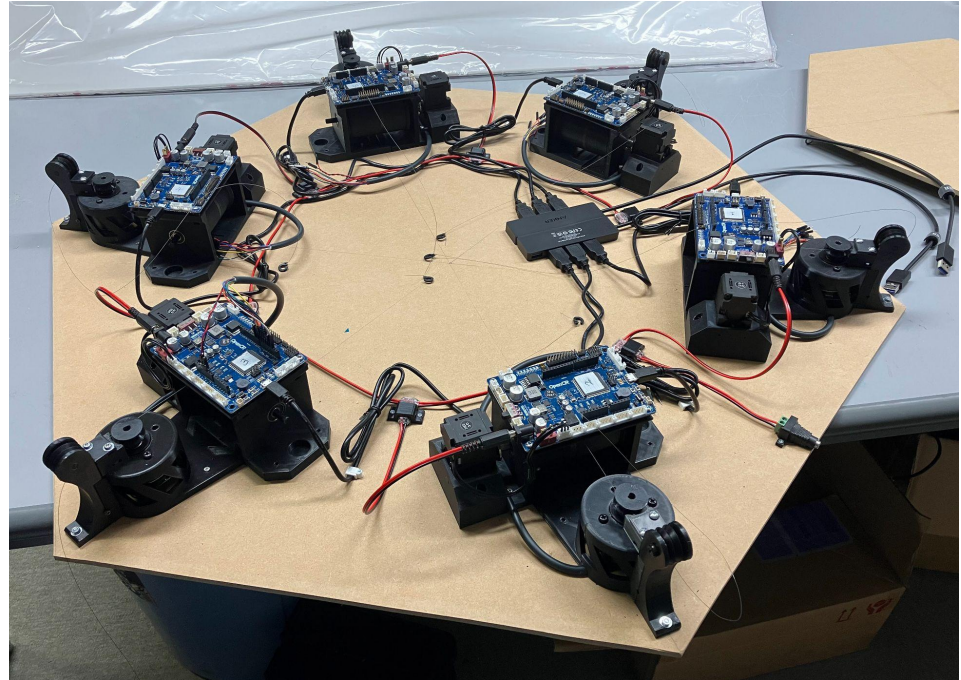
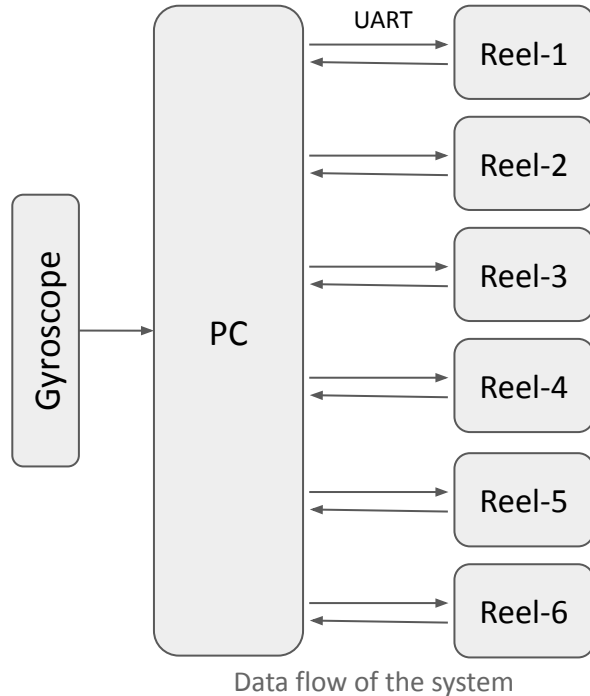
Results for single reel tether movement

- Vertical movement was tested for 5 test cases (200mm, 400mm, 600mm, 800mm, 1000mm).
- 10 trials were conducted for each case.
- Also, the system was test for pushing and pulling the tether.
- Overall average is -0.75mm



Complete assembly

- All modules were connected to the PC using USB hub and can be controlled individually.



Complete assembly

Control interface

- This program scans all connected serial ports and assigns reel ID hard coded in the processing board.
- All six reels can be controlled individually and all together from the GUI.

Parallel Tether Controller Interface

All

Re-assign

☒ Reel 1	Port (COM5)	Torque: On	Verified
☒ Reel 2	Port (COM21)	Torque: On	Verified
☒ Reel 3	Port (COM22)	Torque: On	Verified
☒ Reel 4	Port (COM28)	Torque: On	Verified
☒ Reel 5	Port (COM27)	Torque: On	Verified
☒ Reel 6	Port (COM26)	Torque: On	Verified

Value

-2000

0

2000

Torque on

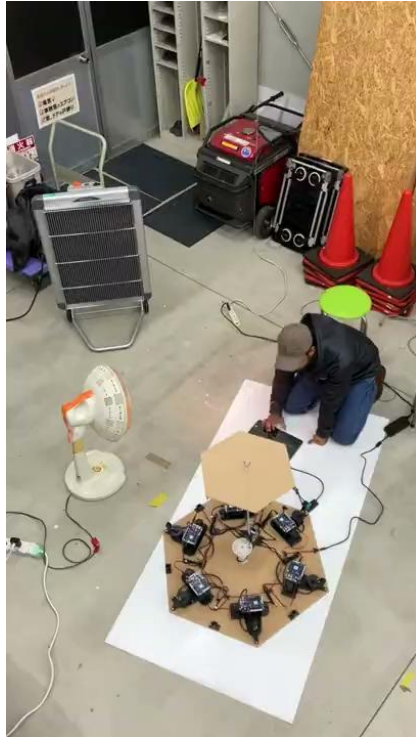
Release

Set length

Control interface (web-based)

Vertical movement test

- Developed system was tested for vertical movements first



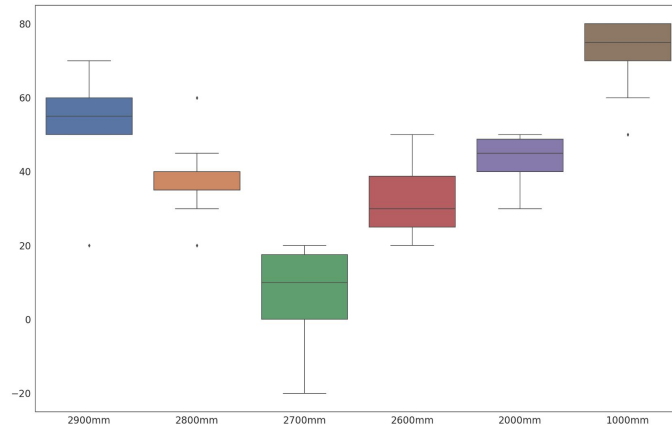
Horizontal movement test

- Developed system was tested for horizontal movements as well

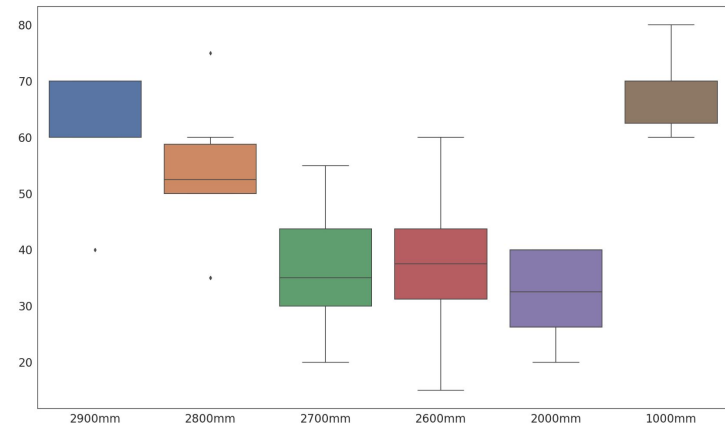


Results for vertical movement

- Vertical movement was tested for 6 test cases (1000mm, 2000mm, 2600mm, 2700mm, 2800mm, 2900mm).
- 10 trials were conducted for each case.
- Also, the system was test for pushing and pulling the tether.
- Overall average is 44.6mm.



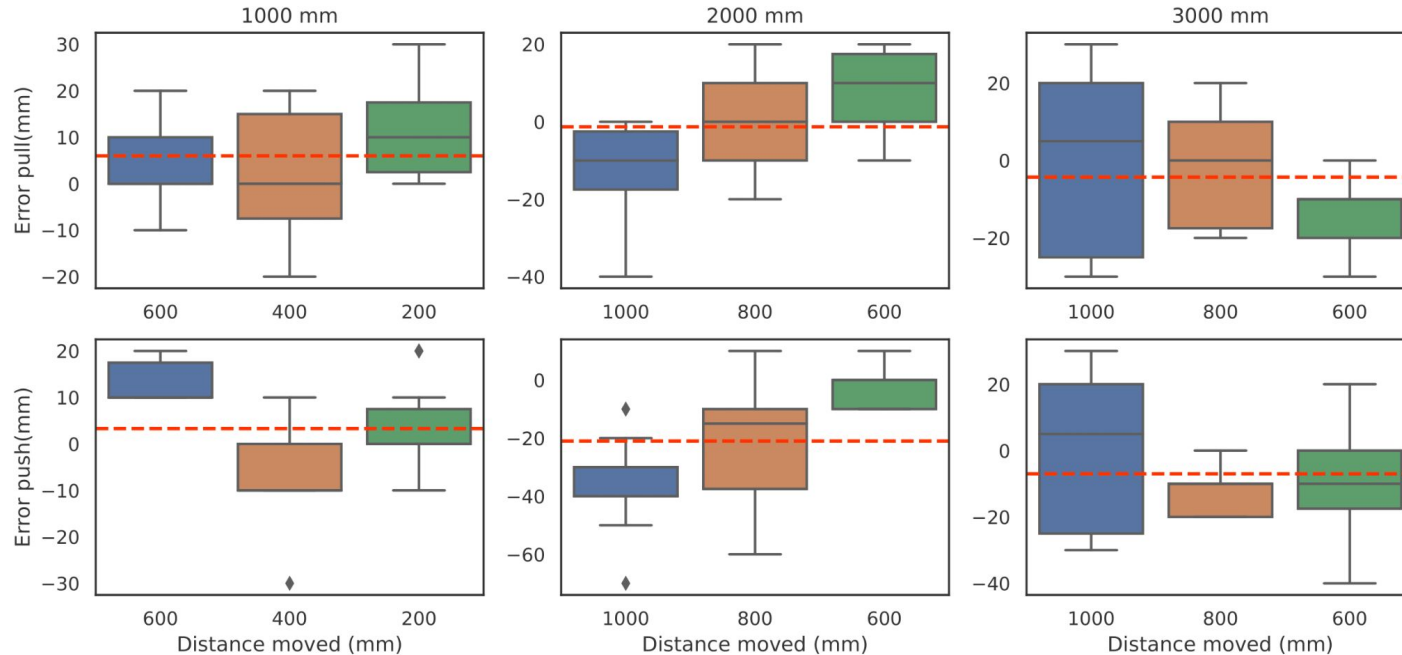
Vertical movement accuracy (pull)



Vertical movement accuracy (push)

Results for horizontal movement

- Horizontal movement was tested for 9 test cases (3 horizontal distances for 3 different height levels).
- Also, the system was test for pushing and pulling the tether.
- 10 trials were conducted for each case.
- Average lies close to 0 for most of the cases except 2000mm pushing case.



Conclusion and future development

- According to the experimental results, this system is capable of collecting data from higher places.
- Issues
 - Tension of tethers significantly reduced in horizontal movement.
 - Difficult to set initial orientation of the sensor module.
 - Size of the helium balloon will be bigger to make sufficient upward force.
- Possible updates
 - Tension of tethers can be adjust automatically using tension sensors.
 - Gyroscope data can be used to automatically adjust moving platform.
 - Inflatable balloon can be used to change the size situationally.

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Thank you.